Theory of Production and Cost

1. Types of Capital Examples:

- **Fixed Capital:** Tools, machines, buildings.
- **Circulating Capital:** Seeds, fuel, raw materials.
- **Real Capital:** Buildings, plants, machinery.
- **Human Capital:** Education, training, skills.
- **Tangible Capital:** Land, vehicles, equipment.
- **Intangible Capital:** Copyrights, goodwill, patents.

2. Cobb-Douglas Production Function:

Formula:

$Q = KL^aC^{(1-a)}$ (Old)

$Q = KL^aC^b$ (New)

Where 'Q' is output, 'L' the quantity of labour and 'C' the quantity of capital. 'K' and 'a' are positive constants.

3. The Law of Variable Proportions or The Law of Diminishing Returns Examples:

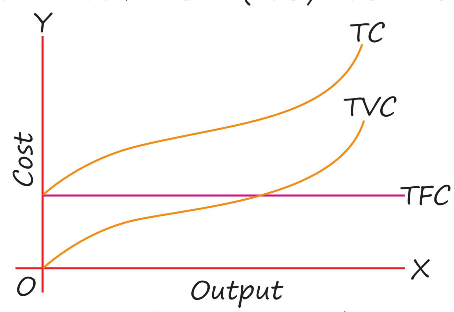
- **Inflection point:** The inflection point is where Total Product (TP) shifts from increasing at an increasing rate to increasing at a decreasing rate. It marks the end of Stage I and the beginning of Stage II in production.
- **Stage of Operation:** A rational producer operates in Stage II, as MP and AP are diminishing but still positive.

4. Capital Formation:

- Process of increasing real capital stock.
- Also called investment.
- **Stages of Capital Formation:**
  1. Savings
  2. Mobilisation of Savings,
  3. Investment

5. Short Run Total Costs:

- Total Cost (TC): TC = TFC + TVC
- Total Variable Cost (TVC): TVC = TC – TFC
- Total Fixed Cost (TFC): TFC = TC – TVC

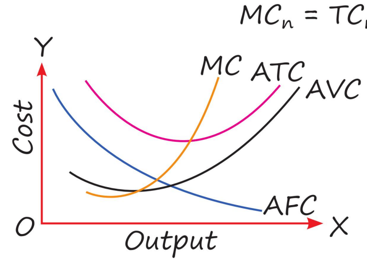


Short Run Total Cost Curves

6. Short Run Average Costs:

- Average Fixed Cost (AFC): AFC = $\frac{\text{ü}}{\quad}$
- Average Variable Cost (AVC): AVC = $\frac{\text{ü}}{\quad}$
- AC = AFC + AVC or AC = $\frac{\text{ü}}{\quad}$
- Marginal Cost (MC): MC = $\frac{\text{ü}}{\quad}$ or

  $MC_n = TC_n - TC_{n-1}$



Short Run Average Cost Curves

7. Long Run Average Cost Curve (Also known as):

- Planning Curve
- Envelope Curve